

Loan App Concept – Developer Brief

1. Product Overview

We are building a web-based lending platform that provides small short-term loans (R700–R1,000 initially) to salaried employees.

Repayment is done via payroll deduction (preferred) or debit order (backup).

The system must be:

- NCR compliant (South Africa)
- Role-based (Employee, Employer, Admin, Investor)
- Secure and scalable
- Audit-ready

2. Core Objective of the System

The platform must:

- Allow employees to apply for loans online
- Run affordability checks
- Allow employer verification
- Disburse loans
- Track repayment
- Manage loan tiers
- Log all activity for compliance

This is a Loan Management System (LMS).

3. User Roles

Employee:

- Register / login
- Upload ID + payslip
- Apply for loan
- View loan status
- Track repayment history
- View credit tier level

Employer / HR:

- Confirm employment
- Approve payroll deduction
- View employee loan exposure

Admin:

- Approve / reject loans
- Monitor loan book
- Track defaults
- Generate compliance reports
- Manage tiers

Investor (Phase 2):

- View capital deployed
- Track ROI
- Monitor loan performance

4. Loan Logic

Initial Rules:

- Loan cap: R700–R1,000
- One active loan per employee
- Repayment within one salary cycle
- Affordability rule based on net salary
- No new loan if previous unpaid

Tier System:

- Tier 1: New borrower
- Tier 2: After successful repayment
- Tier 3: Higher limit
- Tier 4: Premium borrower

Tier updates automatically after repayment.

5. Core Features Required

Authentication & Security:

- JWT authentication
- Role-based access control
- Encrypted document storage
- POPIA compliance

Loan Application Engine:

- Loan submission
- Automated affordability calculation
- Admin approval option
- Loan agreement generation (PDF)

Repayment Engine:

- Payroll deduction tracking

- Late payment flagging
- Default tracking

Dashboard Analytics:

- Loan book value
- Active loans
- Default rate
- Revenue tracking

Audit Logs:

- All actions logged for compliance

6. Suggested Tech Stack

Frontend: React (mobile-first)

Backend: Node.js + Express

Database: PostgreSQL

Storage: AWS S3

Deployment: AWS / Render / Railway

7. Basic Database Structure

Tables:

- Users
- Companies
- Loans
- Repayments
- Tiers
- Investors
- AuditLogs
- Documents

Each loan links to user, company, repayment schedule, tier level, and status.

8. Workflow Summary

Employee applies

- System runs affordability check
- Employer verifies employment
- Admin approves
- Loan disbursed
- Payroll deduction triggered

- Repayment tracked
- Tier updated
- Investor dashboard updated

9. MVP Scope (Version 1)

Focus on:

- Employee portal
- Admin dashboard
- Basic loan approval
- Manual payroll tracking
- Tier logic
- Compliance logging

Employer & investor portals in Phase 2.

10. End Goal

A scalable, compliant payroll-integrated lending system that can:

- Handle thousands of employees
- Support multiple companies
- Onboard investors
- Expand loan limits safely